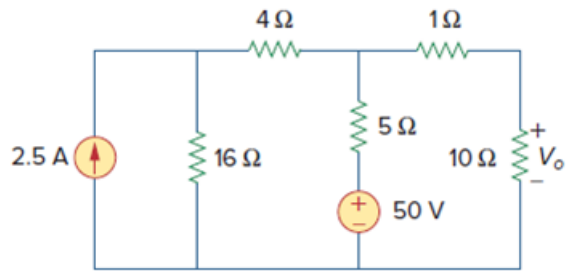


Problem

Apply Thevenin's theorem to find V_o in the circuit of Fig. 4.105.

Figure. 4.105:



Step-by-step solution

Step 1 of 7

Refer to circuit diagram in Figure 4.105 in the text book.

To calculate the Thevenin's Voltage, remove the $10\ \Omega$ resistor. to calculate the as shown in Figure 1.

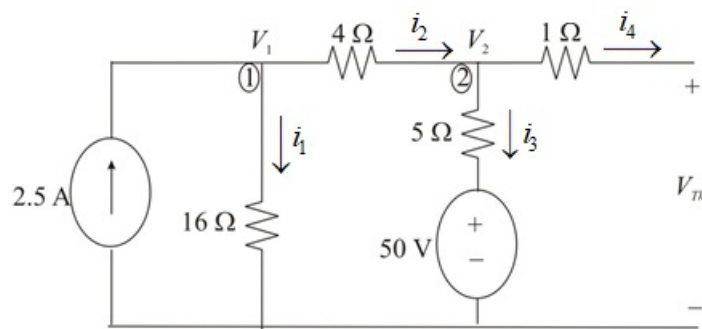


Figure 1

Step 2 of 7

Apply Kirchhoff's Current Law at node 1.

Sum of the currents entering at node 1 is equal to sum of the currents leaving the node 1.

$$2.5 = i_1 + i_2$$

Rearrange the terms.

$$i_1 + i_2 - 2.5 = 0$$

Substitute $\frac{V_1}{16}$ for i_1 and $\frac{V_1 - V_2}{4}$ for i_2 in the equation using ohm's law.

$$\frac{V_1}{16} + \frac{V_1 - V_2}{4} - 2.5 = 0$$

$$\frac{V_1 + 4V_1 - 4V_2 - 40}{16} = 0$$

$$V_1 + 4V_1 - 4V_2 - 40 = 0$$

$$5V_1 - 4V_2 - 40 = 0$$

$$5V_1 = 4V_2 + 40$$

$$V_1 = \frac{4V_2 + 40}{5} \dots\dots (1)$$

Step 3 of 7

Sum of the currents entering at node 2 is equal to sum of the currents leaving the node 2.

Apply Kirchhoff's Current Law at node 2.

$$i_2 = i_3 + i_4$$

Current flows through the $1\ \Omega$ resistor is zero due to open circuit at the output terminals.

So, $i_4 = 0$ and the equation becomes:

$$i_2 = i_3$$

Substitute $\frac{V_2 - 50}{5}$ for i_3 and $\frac{V_1 - V_2}{4}$ for i_2 in the equation using ohm's law.

$$\frac{V_1 - V_2}{4} = \frac{V_2 - 50}{5}$$

Rearrange the terms.

$$\frac{V_2 - 50}{5} + \frac{V_2 - V_1}{4} = 0$$

$$\frac{4V_2 - 200 + 5V_2 - 5V_1}{20} = 0$$

$$9V_2 - 200 - 5V_1 = 0$$

Substitute $\frac{4V_2 + 40}{5}$ for V_1 .

$$9V_2 - 200 - 5\left(\frac{4V_2 + 40}{5}\right) = 0$$

$$9V_2 - 200 - 4V_2 - 40 = 0$$

$$5V_2 - 240 = 0$$

$$5V_2 = 240$$

$$V_2 = \frac{240}{5}$$

$$= 48\text{ V}$$

Here, the voltage V_2 represents the Thevenin's voltage V_{Th} .

$$V_{Th} = 48\text{ V}$$

Therefore, the value of the voltage V_{Th} is 48 V.

To calculate the Thevenin's resistance, open circuit the current source and short circuit the voltage source. The modified circuit is shown in Figure 2.

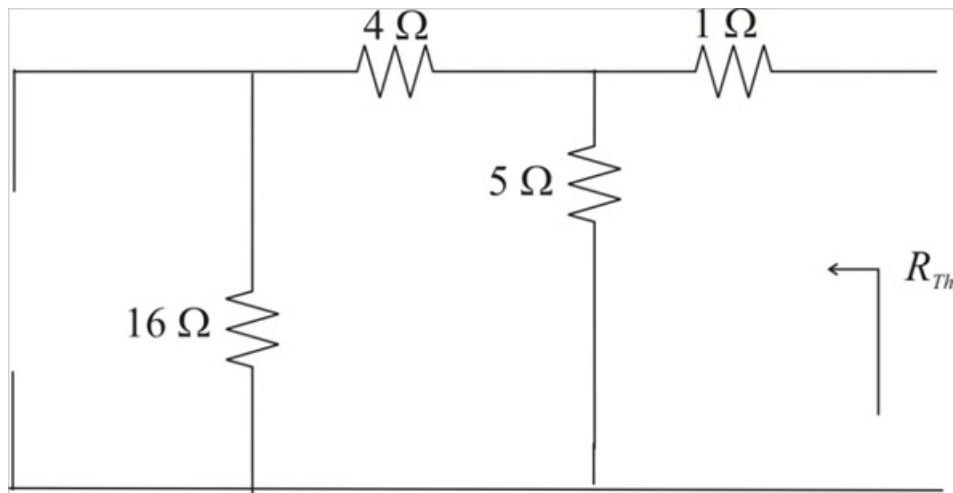


Figure 2

Step 5 of 7

Calculate the value of the Thevenin's resistance R_{Th} .

$$\begin{aligned}
 R_{Th} &= 1 + [5 \parallel (4 + 16)] \\
 &= 1 + [5 \parallel 20] \\
 &= 1 + \frac{(5)(20)}{5 + 20} \\
 &= 1 + \frac{100}{25} \\
 &= 1 + 4 \\
 &= 5\ \Omega
 \end{aligned}$$

Therefore, the value of the Thevenin's resistance R_{Th} is, $5\ \Omega$.

Step 6 of 7

The Thevenin's equivalent circuit is shown in Figure 3.

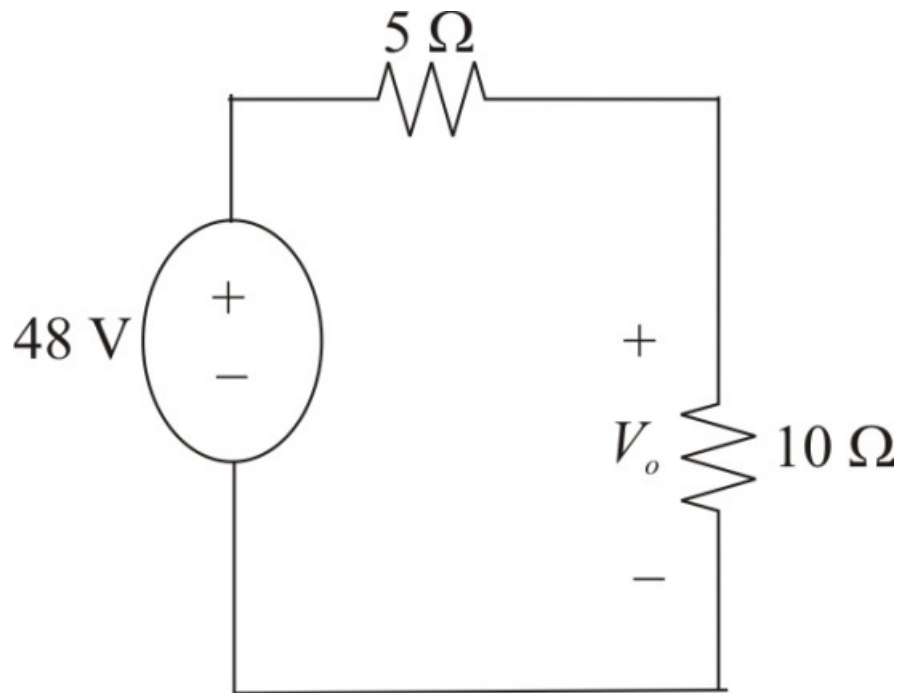


Figure 3

Step 7 of 7

Calculate the value of the output voltage V_o using voltage division rule.

$$\begin{aligned} V_o &= (48) \left(\frac{10}{5+10} \right) \\ &= (48) \left(\frac{10}{15} \right) \\ &= (48) \left(\frac{1}{3} \right) \\ &= 16 \text{ V} \end{aligned}$$

Therefore, the value of the output voltage V_o is, 16 V.

[Comments \(2\)](#)



Anonymous

10/15 doesn't equal 1/3



Anonymous

32 V not 16 v